Claim Correction and Artifact Growth in AI-Assisted Theoretical Research: A Descriptive Longitudinal Case Study

29 October 2026

Abstract

We study two preserved artifact states of an AI-assisted foundational-physics project: an initial scalar boundary-moment programme and a continuation that inherited its manuscripts, calculations, errors and feedback. The later state adds 580 relative paths. A saved hash comparison records 208 identical and eight modified baseline paths, with none missing. Five baseline project manuscripts total 85 pages. The continuation introduced positive noncommutative boundary algebras, exact finite operational calibrations, source-specialized local gauge benchmarks and explicit correction records. These are heterogeneous scientific outputs and cannot be measured by file growth alone. We reproduce the published filename classification on 315 new root Markdown files: its selected fraction is 51.43% of files and 56.35% of lines. Removing the ambiguous term MATRIX changes the file fraction to 49.84% and the line fraction to 54.84%. Thus a majority-of-files interpretation is not robust to one plausible rule change. We reconstruct representative mathematical and encoding corrections, distinguish computation from analytic and physical claims, and propose a compact evidence-routing protocol. The study is descriptive: inherited context, unequal durations, changing tools and human feedback prevent causal attribution to a model family or audit policy. The released metadata and offline script reproduce the numerical summaries without access to private conversations, model providers or the original research directories.

1 Research question and contribution

Long theoretical projects produce more than equations. They produce conjectures, code, source comparisons, rejected arguments, manuscripts and records of their own status. Language-model agents can accelerate all of these activities. Consequently, a growing archive can contain genuine local progress and growing maintenance burden at the same time. The relevant empirical question is what the retained artifacts support, rather than whether their volume sounds like a successful research programme.

This paper asks three descriptive questions. How much of a later corpus is inherited from an earlier state? Which concrete corrections changed the scientific claim boundary? How sensitive is a simple quantitative description of control-related prose to its classification rule? The unit of observation is an artifact or claim, not a model invocation. We do not estimate the causal effect of an audit intervention or rank foundation models.

The scientific setting is a boundary-completion approach to quantum gravity. Its early primitive was a commuting family of thermal quantities $Z(\beta)$ and their positive moments. The continuation broadened this to noncommutative boundary algebras, physical reductions, reflection structures and completely positive operations. This background makes the distinction between a local mathematical result and an integrated physical theory especially

consequential. It also provides unusually explicit examples of corrections: an incorrect quotient type, an insufficient positivity test, an overinterpreted finite model and a mismatch between a mathematical projection and its code encoder.

Our quantitative contribution is a reproducible reaggregation and sensitivity calculation on preserved metadata. Our qualitative contribution is a claim-level case reconstruction, with each example tied to an actual mathematical distinction. The resulting protocol is a research-method proposal. It is not an empirically validated optimization algorithm for science.

2 Materials and chronology

2.1 Two artifact states

The initial state contains 216 files and the continuation state 796 at the declared operational cutoff, 11 July 2026 at 20:37:59 Beijing time. The saved manifest was generated on 12 July by selecting files using mutable filesystem modification times. It is therefore a reconstructed historical dataset, not a cryptographically frozen log created at the cutoff. Hashes identify the included bytes but cannot establish when those bytes were first produced. The present paper and its portability calculation were prepared on 29 October.

The continuation inherited the initial manuscripts, code, literature, conjectures, failed routes and human criticism. It ran for a different duration with different tools and task decomposition. These are major design confounders, not small statistical adjustments. There is no randomized or matched comparison, no no-audit counterfactual and no complete transcript-level allocation of work to model versions.

Table 1: Preserved corpus aggregates. Counts refer to the declared reconstructed states and are not an estimate of scientific correctness.

Quantity	Initial state	Continuation
Files	216	796
Bytes	6,864,474	16,731,631
Markdown files	23	358
Markdown lines	4,462	85,298
Python sources	3	40
Wolfram-language sources	116	116
TeX sources	6	23
PDF files	7	42

The baseline has five project manuscripts, with page counts 25, 21, 14, 12 and 13, totaling 85. They concern a synthesis, thermal moments, projection kernels, near-extremal black holes and Gaussian chaos. The fifth PDF is in a track-specific paper directory, rather than a fifth numbered paper folder. This corrects a path description in the earlier evidence map without changing the manuscript count. Two other PDFs in the initial total are external literature, so the seven-PDF count is not seven original project papers. The page counts were checked against the current preserved files with hashes matching the saved manifest during the preceding recovery. The portable metadata records that check; it does not redistribute those PDFs or re-run a page parser on unavailable source files.

2.2 Released evidence and excluded material

The package contains the byte-preserved 315-row root-filename classification table, aggregate counts, a reduced path/hash relation table, the five manuscript page-count records and an offline reproduction script. Internal source names are confined to provenance metadata. The reduced relation table replaces each relative path by its SHA-256 identifier and retains baseline and continuation content hashes and their relation. It permits independent recounting of the saved comparison while avoiding redistribution of source contents.

Private conversations, browser data, credentials, model-provider records and third-party full papers are excluded. This limits fine-grained attribution and behavioral inference. It does not prevent reaggregation of the released tables or inspection of the mathematical distinctions below. Reproducible computation and broader scientific replicability are distinct obligations [7, 9].

3 Methods

3.1 Relative-path and content comparison

Let B and C be the included relative-path sets for the initial and later states, and let $h_B(p), h_C(p)$ be their content hashes. We count

$$N_{=} = \#\{p \in B \cap C : h_B(p) = h_C(p)\}, \tag{1}$$

$$N_{\neq} = \#\{p \in B \cap C : h_B(p) \neq h_C(p)\}, \tag{2}$$

$$N_{-} = \#(B \setminus C), \qquad N_{+} = \#(C \setminus B). \tag{3}$$

The saved comparison gives $(N_{=}, N_{\neq}, N_{-}, N_{+}) = (208, 8, 0, 580)$. The eight changes comprise two root Markdown files and six source/PDF/build artifacts of the projection-kernel paper. Thus the later state is largely additive at the path level. This does not establish independent rediscovery, semantic improvement of every unchanged file, or attribution of each addition.

3.2 Filename-rule classification

The published rule selects a new root Markdown file if its filename contains at least one declared token, case-insensitively. The tokens are AUDIT, CHECKPOINT, LEDGER, MATRIX, ROADMAP, READINESS, MAINLINE, CONSTITUTION, SUMMARY, HANDOFF, WORK-SESSION, CONTINUATION-LOG, NIGHT-RUN, FILE-MAP, DEPENDENCY-GRAPH, MATURITY, SANITY-CHECK, CRITERION, TARGET, OBLIGATION, GATE, CANDIDATE, PROTOCOL, POLICY, PROGRAM and SCHEMA. Matches are substring matches, not a semantic annotation of document contents.

For file i, let ℓ_i be its preserved line count and $s_i(T)$ the indicator that its filename matches token set T. We report

$$f(T) = \frac{\sum_i s_i(T)}{315}, \qquad g(T) = \frac{\sum_i \ell_i s_i(T)}{77,375}. \tag{4}$$

The denominator is fixed across variants. These are deterministic population summaries of the preserved table. Sampling confidence intervals would not address their principal uncertainty, which is semantic validity of the rule.

We prespecify two transparent sensitivity variants for this reanalysis: remove MATRIX alone; then also remove MAINLINE and CONSTITUTION. MATRIX can name an actual matrix theorem as readily as a control matrix. The second variant tests a broader group of content-bearing labels. Neither is offered as a ground-truth classification. Removing ambiguous tokens can remove real control documents as well as substantive mathematics.

3.3 Claim-level reconstruction

For each qualitative example we distinguish five things: the object actually constructed, the assumptions, the evidence channel, the claimed extension and the correction. Direct algebra can decide a sign or finite counterexample; it cannot decide an unencoded analytic premise. A source theorem contributes only under its hypotheses and convention map. An AI critic's agreement is a record of further AI review, not human expert review.

The examples were chosen because the retained record contains concrete artifacts and identifiable corrections, rather than sampled randomly from all research tasks. This is purposive case selection. It supports explanation of possible failure mechanisms, not an estimate of their frequency across AI research projects.

4 Quantitative results

4.1 Inheritance and growth

There are 335 new Markdown files containing 80,807 lines. Of these, 315 files and 77,375 lines occur at root level. The remaining 20 new Markdown files contain 3,432 lines. The Python source count increases by 37 while the Wolfram-source count is unchanged. These numbers describe artifact allocation and inherited structure. A source count is not a count of correct calculations: one program may contain many identities, and multiple programs may test the same identity.

4.2 Sensitivity of the majority headline

Table 2 gives the reproduced results. Five filenames match only MATRIX, accounting for

Table 2: Sensitivity on the same 315-row preserved table. The original rule-selected majority of files disappears when MATRIX alone is removed.

Rule	Files	File %	Lines	Line %
Original tokens	162	51.43	43,603	56.35
Without MATRIX	157	49.84	42,433	54.84
Also without MAINLINE, CONSTITUTION	154	48.89	39,981	51.67

1,170 lines. They include a full nonminimal Green-matrix theorem and a finite regulator matrix-sector theorem as well as control/selection matrices. The original arithmetic is correct; its interpretation as the fraction of control-only material is not secured by that arithmetic. In particular, it would be false to call 51.43% an estimated waste fraction. The larger line fraction also does not establish that control activity caused weaker research output.

The conclusion is bounded and useful: report the exact rule-selected subset and its sensitivity, rather than relying on a binary majority headline. Determining actual semantic allocation would require a versioned annotation scheme, independent document-level coding and an agreement analysis. Those data are not present, and no such coding was invented for this study.

5 Mathematical and encoding corrections

5.1 A commuting observable family is not a full boundary algebra

An initial thermal family gives mixed moments

$$M_n(\beta_1, \ldots, \beta_n) = \omega\big(Z(\beta_1) \cdots Z(\beta_n)\big). \tag{5}$$

Even all such moments concern the commutative algebra generated by those observables. They do not supply ordered insertions outside that algebra, constraint reduction or sewing operations. The continuation's noncommutative boundary formulation is therefore a substantive correction of the scope of the primitive, not merely a new name.

A minimal example makes the limitation concrete. The qubit states $\rho_\pm = |\pm\rangle\langle\pm|$ agree on every diagonal observable, but

$$\mathrm{Tr}((\rho_+ - \rho_-)X) = 2, \qquad \tfrac{1}{2}\|\rho_+ - \rho_-\|_1 = 1. \tag{6}$$

Restricted ambiguity can disappear when the measurement algebra is enlarged. To interpret it physically one must also show that both states satisfy the same constraints, dynamics and limit conditions. The continuation consequently rephrased completion ambiguity as noninjectivity of restriction on a declared physical state set. This is a framework clarification, not a construction of a gravitational landscape.

5.2 Scalar sewing positivity is weaker than complete positivity

For a positive map $\Phi : A \to B$ and positive scalar functional ρ,

$$\sum_{ij} \bar{c}_i c_j \rho(\Phi(a_i^* a_j)) = \rho\big(\Phi(x^* x)\big) \geq 0, \qquad x = \sum_j c_j a_j. \tag{7}$$

Thus every scalar sewing Gram is positive without assuming complete positivity. Even all such scalar tests do not certify CP: transposition on M_2 passes them but has a Choi matrix with spectrum $(-1, 1, 1, 1)$. The correct stronger test uses operator-valued matrices $[\Phi(a_i^* a_j)] \geq 0$ at every size, or an equivalent finite Choi/Kraus certificate where applicable [2, 12]. This correction is methodologically instructive because adding more tests of the wrong scalar kind cannot close the missing ancillary condition.

The finite operational constructions in the corpus do use actual Kraus and isometry arguments. Their CP claims survive this clarification. What changes is the interpretation of a weaker diagnostic, not every positive result associated with it. Earlier positive boundary quantum formalisms remain relevant prior art [8].

5.3 Two distinguishability questions

A two-outcome finite instrument yields common-fiber posterior densities ρ_0, ρ_1 with trace distance $|2\lambda - 1|$ in its selected example. If the classical outcome record is retained, the conditioned states live in different summands of a direct-sum algebra and have distance 1. Forgetting the record and identifying the quantum fibers is a separate channel. The earlier correction changed a statement about the operational object, not the numerical tolerance. This is a case where a correct decimal value was attached to the wrong comparison space.

5.4 Compression needs its declared algebra

Compression $E(a) = V^* a V$ by an isometry is UCP. It need not be multiplicative. For compression of M_2 onto its first coordinate,

$$E(e_{12})E(e_{21}) = 0, \qquad E(e_{12}e_{21}) = E(e_{11}) = 1. \tag{8}$$

The finite matched-charge construction is a star quotient because its observable algebra is a block-diagonal direct sum and compression deletes whole unmatched summands. Broadening that algebra to arbitrary cross-charge operators invalidates the quotient conclusion. A theorem on one declared algebra is not automatically a theorem on a larger, more physically suggestive one.

5.5 Code can verify the wrong projection

An interaction analysis attempted to bound source degrees after removing a nonminimal BRST quartet. The intended gauge-fixed minimal projection has the conjugated form

$$\Pi_{\mathrm{gf}} = C_{\mathrm{gf}}\, \iota\pi\, C_{\mathrm{gf}}^{-1}. \tag{9}$$

It is not generally naive deletion of nonminimal variables. An enumerator using deletion rules can be correct for its encoded monomials while failing to establish a degree bound under (9). Repeating that enumeration cannot repair the model-to-code mismatch. The corrected benchmark therefore excludes the proposed source-filtration result and leaves the actual anomaly-map image uncomputed.

The valid general quartet contraction is established BRST mathematics [1]. The useful project increment lies in its restricted, support/normalization-aware specialization and explicit candidate primitives, not an unsupported declaration that an interacting anomaly vanishes.

5.6 A sourced benchmark does not derive its input dynamics

The local gauge construction starts from an Einstein–scalar action and derives compatible free Green matrices, normalization maps and BRST algebraic conditions. These can be real technical achievements even though the target action is supplied. They answer a compatibility question. They do not show that a boundary-completion principle generated Einstein gravity.

The physical-state existence theorem also must be distinguished from the surrounding functional algebra. A positive physical Hadamard state does not by itself imply a normalized boundary instrument, descent of an ordinary dagger to the same BRST complex, all-equicausal time ordering or an interacting quantum master equation. The later benchmark preserves these separate limits. This is an example of successful scope correction with a substantial retained technical module.

5.7 Decaying correlations do not prove information destruction

One historical synthesis associated continuous spectral dephasing with information loss. On $L^2([0,1], dx)$, however,

$$(U_t f)(x) = e^{-itx} f(x) \tag{10}$$

is a strongly continuous unitary group, while the constant-vector correlation is $(1 - e^{-it})/(it)$ and tends to zero. A spectral recurrence distinction is not a distinction between unitary and nonunitary dynamics. Neither this counterexample nor a finite Page-shaped spectral-form-factor proxy decides black-hole radiation entropy. The correction removes a physical inference without invalidating the elementary spectral calculations.

6 What can be concluded about the workflow?

The record supports a mixed account. The continuation produced concrete scope corrections, exact finite calibrations and local sourced benchmarks. It also produced many mirrored status

and obligation documents. The latter can increase synchronization burden: if every claim is copied into every manual mirror, a simple bookkeeping model has cost proportional to $|\mathcal{C}||\mathcal{M}|$. This is a conditional cost model, not a fitted law or measured runtime effect.

We use *audit inflation* as an interpretive hypothesis: control artifacts may grow without a comparable increase in independently closed claims. The case provides examples of duplication and repeated deferral. It does not provide a validated measure of closed scientific content at each time, nor the counterfactual output under a different audit policy. Therefore it does not establish that auditing reduced productivity. Audits also detected real errors; the same examples can support beneficial error correction and costly maintenance simultaneously.

An agent dialogue can reveal a sign error or hidden assumption, but role separation is not evidential independence. Models may share project context and the same unexamined premise. A fresh implementation, primary-source passage, proof-assistant kernel or external observation changes the evidence channel. These channels have different scopes: formal verification certifies the formal statement, not its physical interpretation.

The comparison is likewise insufficient for model-capability ranking. Later artifacts benefited from inherited work and corrective feedback; earlier artifacts were produced under different circumstances. Statements that one stage is better in a specified artifact property can be supported. A generic claim that its model family is better at foundational physics cannot.

7 Relation to other AI research systems

Program-search systems such as FunSearch combine generated programs with an executable evaluator [11]; theorem-proving systems can obtain kernel-level feedback [10]. End-to-end research systems automate larger portions of ideation, experimentation and writing [5], while Co-Scientist studies a hypothesis-generation and refinement workflow [3]. These are useful contrasts in the available feedback channels. Their reported results do not estimate the effect of the workflow studied here.

For the historical literature context we pin the December 2025 first version of PhysMaster [6]. Its later August version changes the author list and evaluation description. Substituting current metadata into a July-cutoff narrative would conflate versions. The June 2026 physical-science multi-agent benchmark also predates the cutoff [4]. This is an illustrative primary-source comparison, not a systematic review, performance replication or common-benchmark ranking.

8 A compact evidence-routing protocol

The cases motivate a small release structure. First state the central scientific question and the strongest claim that answers it. Give each claim a type—definition, standard input, derived theorem, finite computation, conditional application, conjecture or refuted implication—and record its exact object and assumptions. A finite calibration and a physical continuum construction should not share an unqualified verification label.

Second, identify the encoding bridge. For a program, state which mathematical objects become arrays or monomials and why the encoded operations represent the intended map. For a source theorem, state the hypotheses and convention transport. A correct program or citation cannot compensate for a missing bridge. The scalar-versus-amplified positivity example shows why test strength must be specified, not merely the number of successful tests.

Third, preserve one current machine-readable claim table and versioned history, rather than manually copying status across many active mirrors. One short release report should state the strongest retained result, the strongest withdrawn claim, evidence actually rerun and

material gaps. This proposal follows the observed failure modes; its productivity benefit has not been measured in this case.

Finally, choose the next research operation by its scientific consequence. A bounded proof or counterexample can resolve a question. Another criterion document is useful only if it changes an acceptance decision or executable test. A failed parent programme can still leave an independent framework, method, benchmark or mathematical result. Requiring complete quantum gravity before recognizing those outputs would discard the very distinctions this case makes observable.

9 Limitations and conclusion

The study has one project, two reconstructed artifact states, purposively selected qualitative cases and incomplete production provenance. File timestamps are mutable, filename categories are semantic proxies, and the technical modules have received AI review rather than human expert review. Some analytic proof chains remain source-specialized and have not received a fresh complete independent audit in this paper. No intervention effect, general model ranking, waste percentage or comprehensive originality certificate follows.

Within those limits, the numerical findings are exact properties of released metadata: substantial inheritance, 580 additional paths, five baseline project manuscripts totaling 85 pages, and a majority-of-files interpretation that changes under removal of MATRIX. The mathematical cases identify concrete ways fluent research can outgrow its evidence and ways correction can preserve useful results. The resulting lesson is to align a claim with its actual object, evidence channel and promotion boundary. Volume and polish do not perform that alignment automatically.

Workflow and unresolved attribution

The operator reports that earlier work was led mainly by Fable 5, whereas results from July 2026 onward were predominantly produced by GPT models. DeepSeek and multiple GPT models also participated in review, checking and manuscript preparation and organization. This is a reported chronological history, not a complete audited allocation of individual contributions. An earlier result is not reattributed merely because it was revised or packaged later. Human author names and affiliations are intentionally blank. The results have not undergone human expert review. Automated checks and AI reviews are further AI review, not human peer review. Historical source dates, this preparation date and demonstrable public disclosure dates are distinct. No authorship, ownership, publication-priority or journal-acceptance decision is made. The operator states:

> I genuinely do not know how authorship, intellectual credit, or ownership should be attributed for these results.

Data and code availability

The accompanying packet includes the reduced corpus metadata and `reproduce.py`. The default calculation needs only Python's standard library and local files. It reproduces the hash-relation counts, line totals and token sensitivities; it does not claim to recover historical model sessions or reconstruct the old filesystem from present modification times. No raw conversations, secrets or third-party full papers are required.

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